

Introduction to Probability Statistics and Random Processes - Formulas

Indhold

Chapter 1 - Basic Concepts	5
De Morgan's law	5
Distributive law	5
Axioms of Probability	5
Inclusion-exclusion principle	5
Conditional probability of A given B	5
Chain rule for conditional probability	5
Independence	6
Law of Total Probability	6
Bayes' Rule	6
Chapter 2 - Combinatorics: Counting Methods	7
k -permutations of n distinguishable objects	7
The number of k -combinations of an n -element set	7
Binomial Formula	7
Chapter 3 - Discrete Random Variables	8
Range	8
Probability mass function (PMF)	8
Independent discrete random variables	8
Cumulative Distribution Function (CDF)	8
Expectation	8
Expected value of a function of a random variable (LOTUS)	9
Variance	9
Standard deviation	9
Chapter 4 - Continuous and Mixed Random Variables	10
Definition on a continuous random variable	10
Range	10
Expected Value	10
LOTUS	10
Variance	11

Theorem 4.1	11
Chapter 5 - Joint Distributions: Two Random Variables	12
Two Discrete Random Variables	12
Joint Probability Mass Function (PMF)	12
Marginal PMFs	12
Joint Cumulative Distribution Function (CDF)	12
Conditional PMF/CDF given event A	12
Conditional PMF	13
Independent Random Variables	13
Conditional Expectation	13
Law of Total Probability	14
LOTUS.....	14
Law of iterated Expectations	14
Independent Random Variables	14
law of Total Variance	15
Two Continuous Random Variables (PDF)	15
Joint Probability Density Function.....	15
Marginal PDFs	15
Joint Cumulative Distribution Function (CDF)	16
Continuous random variable and event A	16
Two jointly continuous random variables X and Y	17
Independent Random Variables	17
Law of Total Probability	18
LOTUS for two continuous random variables.....	18
The Method of Transformations	18
Covariance.....	19
Variance of a sum.....	19
Correlation Coefficient	20
Bivariate normal/Jointly normal	20
Chapter 6 - Multiple Random Variables	22
Independent and identically distributed	22
Sums of Random Variables	22
Moment Generating Functions	22
Finding Moments from MGF	23

Sum of Independent Random Variables	23
Correlation and Covariance Matrix	23
Chapter 8 - Statistical Inference I: Classical Methods	24
Random sample	24
Sample mean	24
Theorem 8.1	25
Bias	25
Unbiased	25
Mean squared error (MSE).....	25
Sample variance	26
Sample standard deviation	26
Likelihood function.....	26
Maximum Likelihood Estimator (MLE)	27
Asymptotic Properties of MLEs	27
Interval Estimation	28
Pivotal Quantity	28
Find Confidence interval	28
Chi-Squared Distribution.....	30
The t-Distribution.....	31
Confidence Intervals for the mean of Normal Random Variables	32
Hypothesis testing.....	32
Two-sided hypothesis testing $H_0: \mu = \mu_0, H_1: \mu \neq \mu_0$	33
One-sided hypothesis testing for the mean $H_0: \mu \geq \mu_0, H_1: \mu < \mu_0$	33
One-sided hypothesis testing for the mean $H_0: \mu \leq \mu_0, H_1: \mu > \mu_0$	34
P-value	34
Likelihood Ratio Test for Simple Hypotheses	35
Simple Linear Regression	36
Coefficient of Determination.....	36
Chapter 11 - Some important random processes	37
Markov Chains	37
Discrete-Time Markov Chains	37
Transition probabilities	37
State Transition Matrix	37
State Transition Diagram.....	37

State Probability Distributions	38
The chapman-Kolmogorov equation	38
n -step transition matrix	38
Accessible	38
Communicate.....	38
Irreducible	39
Recurrent and transient state.....	39
Periodic and aperiodic	39
Absorption Probabilities	40
Mean Hitting Time.....	40
Mean Return Times	40
Limiting Distributions.....	41
Finite Markov Chains	41

Chapter 1 - Basic Concepts

De Morgan's law

s. 9

$$(A_1 \cup \dots \cup A_n)^c = A_1^c \cap \dots \cap A_n^c$$

$$(A_1 \cap \dots \cap A_n)^c = A_1^c \cup \dots \cup A_n^c$$

Distributive law

s. 9

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Axioms of Probability

s. 21

Axiom 1: For any event A, $P(A) \geq 0$

Axiom 2: Probability of the sample space S is $P(S) = 1$

Axiom 3: If $A_1, A_2, A_3, \dots$ are disjoint events, then $P(A_1 \cup A_2 \cup A_3 \cup \dots) = P(A_1) + P(A_2) + P(A_3) + \dots$

Inclusion-exclusion principle

s. 27

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

Conditional probability of A given B

s. 40

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad \text{when } P(B) > 0$$

Chain rule for conditional probability

s. 45

$$P(A_1 \cap \dots \cap A_n) = P(A_1)P(A_2|A_1)P(A_3|A_2, A_1) \dots P(A_n|A_{n-1}, A_{n-2} \dots A_1)$$

Independence

s. 46-47 + 49

$$P(A \cap B) = P(A)P(B)$$

A, B and C are independence if all the following conditions hold.

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cap C) = P(A)P(C)$$

$$P(B \cap C) = P(B)P(C)$$

$$P(A \cap B \cap C) = P(A)P(B)P(C)$$

if $A_1 \dots A_n$ are independent then

$$P(A_1 \cup \dots \cup A_n) = 1 - (1 - P(A_1)) \dots (1 - P(A_n))$$

Law of Total Probability

s. 52

If $B_1, B_2, B_3, \dots$ is a partition of the sample space S, then for any event A we have

$$P(A) = \sum_i P(A \cap B_i) = \sum_i P(A|B_i)P(B_i)$$

Bayes' Rule

s. 55

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

Chapter 2 - Combinatorics: Counting Methods

k -permutations of n distinguishable objects

s. 87

$$P_k^n = \frac{n!}{(n-k)!}, \quad \text{for } 0 \leq k \leq n$$

The number of k -combinations of an n -element set

s. 88

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}, \quad \text{for } 0 \leq k \leq n$$

$\binom{n}{k}$ is also call binomial coefficient

Binomial Formula

s. 93

$$P(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Chapter 3 - Discrete Random Variables

Range

s. 109

If X is a discrete random variable, then its range R_X is a countable set:

$$R_X = \{x_1, x_2, x_3, \dots\}$$

Probability mass function (PMF)

s. 109-110

$$P_X(x_k) = P(X = x_k), \quad \text{for } k = 1, 2, 3, \dots$$

$$P_X = \begin{cases} P(X = x) & \text{if } x \text{ in } R_X \\ 0 & \text{otherwise} \end{cases}$$

Independent discrete random variables

s. 113-114

X and Y are independent if

$$P(X = x, Y = y) = P(X = x)P(Y = y), \quad \text{for all } x, y$$

n discrete random variables are independent if

$$P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n) = P(X_1 = x_1)P(X_2 = x_2) \dots P(X_n = x_n), \quad \text{for } x_1, x_2, \dots, x_n$$

Cumulative Distribution Function (CDF)

s. 135 + 137

$$F_X(x) = P(X \leq x), \quad \text{for all } x \in \mathbb{R}$$

$$P(a < X \leq b) = F_X(b) - F_X(a)$$

Expectation

s. 139 + 142

$$EX = \sum_{x_k \in R_X} x_k P(X = x_k) = \sum_{x_k \in R_X} x_k P_X(x_k)$$

Expectation is linear (Linearity of expectation)

$$E[aX + b] = aEX + b$$

$$E[X_1 + X_2 + \dots + X_n] = EX_1 + EX_2 + \dots + EX_n$$

Expected value of a function of a random variable (LOTUS)

s. 144

$$E[g(X)] = \sum_{x_k \in R_X} g(x_k)P_X(x_k)$$

Variance

s. 146-148

$$\text{Var}(X) = E[(X - \mu_X)^2]$$

$$\text{Var}(X) = E[X^2] - [EX]^2$$

Variance is not a linear operator

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

If $X_1, X_2, \dots, X_n$ are independent random variables and $X = X_1 + X_2 + \dots + X_n$, then

$$\text{Var}(X) = \text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n)$$

Standard deviation

s. 146

$$SD(X) = \sigma_X = \sqrt{\text{Var}(X)}$$

Chapter 4 - Continuous and Mixed Random Variables

Definition on a continuous random variable

s. 163-165

A random variable X with CDF $F_X(x)$ is said to be continuous if $F_X(x)$ is a continuous function for all $x \in \mathbb{R}$.

Consider a continuous random variable X with an absolute continuous CDF $F_X(x)$. The function $f_X(x)$ defined by

$$f_X(x) = \frac{dF_X(x)}{dx} = F'_X(x), \quad \text{if } F_X(x) \text{ is differentiable at } x$$

Is called the probability density function (PDF) of X .

With other words:

$$f'(x) = F(x)$$

Following is true for a continuous random variable

1. $f_X(x) \geq 0$ for all $x \in \mathbb{R}$
2. $\int_{-\infty}^{\infty} f_X(u) du = 1$
3. $P(a < X \leq b) = F_X(b) - F_X(a) = \int_a^b f_X(u) du$
4. For a set A , $P(X \in A) = \int_A f_X(u) du$

Range

s. 166

$$R_X = \{x | f_X(x) > 0\}$$

Expected Value

s. 167

$$EX = \int_{-\infty}^{\infty} x f_X(x) dx$$

LOTUS

s. 168

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

Variance

s. 169

$$\text{Var}(X) = E[(X - \mu_X)^2] = \int_{-\infty}^{\infty} (x - \mu_X)^2 f_X(x) dx$$

$$\text{Var}(X) = EX^2 - (EX)^2 = \int_{-\infty}^{\infty} x^2 f_X(x) dx - \mu_X^2$$

Theorem 4.1

s. 173

Suppose that X is a continuous random variable and $g : \mathbb{R} \rightarrow \mathbb{R}$ is a strictly monotonic differentiable function. Let $Y = g(X)$. Then the PDF of Y is given by

$$f_Y(y) = \begin{cases} \frac{f_X(x_1)}{|g'(x_1)|} = f_X(x_1) * \left| \frac{dx_1}{dy} \right|, & \text{where } g(x_1) = y \\ 0 & \text{if } g(x) = y \text{ does not have a solution} \end{cases}$$

Chapter 5 - Joint Distributions: Two Random Variables

Two Discrete Random Variables

Joint Probability Mass Function (PMF)

s. 219

The joint probability mass function of two discrete random variables X and Y is defined as

$$P_{XY}(x, y) = P(X = x, Y = y)$$

Marginal PMFs

s. 221

Marginal PMFs of X and Y

$$P_X(x) = \sum_{y_j \in R_Y} P_{XY}(x, y_j), \quad \text{for any } x \in R_X$$

$$P_Y(y) = \sum_{x_i \in R_X} P_{XY}(x_i, y), \quad \text{for any } y \in R_Y$$

Joint Cumulative Distribution Function (CDF)

s. 223-224

The joint cumulative distribution function of two random variables X and Y is defined as

$$F_{XY}(x, y) = P(X \leq x, Y \leq y)$$

Marginal CDFs of X and Y

$$F_X(x) = F_{XY}(x, \infty) = \lim_{y \rightarrow \infty} F_{XY}(x, y), \quad \text{for any } x$$

$$F_Y(y) = F_{XY}(\infty, y) = \lim_{x \rightarrow \infty} F_{XY}(x, y), \quad \text{for any } y$$

Conditional PMF/CDF given event A

s. 227

For a discrete random variable X and event A, the conditional PMF of X given A is defined as

$$P_{X|A}(x_i) = P(X = x_i|A) = \frac{P(X = x_i \text{ and } A)}{P(A)}, \quad \text{for any } x_i \in R_X$$

Similarly, we define the conditional CDF of X given A as

$$F_{X|A}(x) = P(X \leq x|A)$$

Conditional PMF

s. 228

For discrete random variables X and Y , the **conditional PMFs** of X given Y and vice versa are defined as

$$P_{X|Y}(x_i|y_j) = \frac{P_{XY}(x_i, y_j)}{P_Y(y_j)},$$

$$P_{Y|X}(y_j|x_i) = \frac{P_{XY}(x_i, y_j)}{P_X(x_i)}$$

for any $x_i \in R_X$ and $y_j \in R_Y$.

Independent Random Variables

s. 228

Two discrete random variables X and Y are independent if

$$P_{XY}(x, y) = P_X(x)P_Y(y), \quad \text{for all } x, y.$$

Equivalently, X and Y are independent if

$$F_{XY}(x, y) = F_X(x)F_Y(y), \quad \text{for all } x, y.$$

Conditional Expectation

s. 231

Conditional Expectation of X :

$$E[X|A] = \sum_{x_i \in R_X} x_i P_{X|A}(x_i),$$

$$E[X|Y = y_j] = \sum_{x_i \in R_X} x_i P_{X|Y}(x_i|y_j)$$

Law of Total Probability

s. 233

Law of Total Probability:

$$P(X \in A) = \sum_{y_j \in R_Y} P(X \in A | Y = y_j) P_Y(y_j), \quad \text{for any set } A.$$

Law of Total Expectation:

1. If $B_1, B_2, B_3, \dots$ is a partition of the sample space S ,

$$EX = \sum_i E[X|B_i]P(B_i) \quad (5.3)$$

2. For a random variable X and a discrete random variable Y ,

$$EX = \sum_{y_j \in R_Y} E[X|Y = y_j]P_Y(y_j) \quad (5.4)$$

LOTUS

s. 235

Law of the unconscious statistician (LOTUS) for two discrete random variables:

$$E[g(X, Y)] = \sum_{(x_i, y_j) \in R_{XY}} g(x_i, y_j) P_{XY}(x_i, y_j) \quad (5.5)$$

Law of iterated Expectations

s. 241

Law of Iterated Expectations: $E[X] = E[E[X|Y]]$

Independent Random Variables

s. 242

If X and Y are independent random variables, then

1. $E[X|Y] = EX$;
2. $E[g(X)|Y] = E[g(X)]$;
3. $E[XY] = EXEY$;
4. $E[g(X)h(Y)] = E[g(X)]E[h(Y)]$.

law of Total Variance

s. 244

Law of Total Variance:

$$\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}(E[X|Y]) \quad (5.10)$$

Two Continuous Random Variables (PDF)

Joint Probability Density Function

s. 257

Definition 5.1

Two random variables X and Y are **jointly continuous** if there exists a nonnegative function $f_{XY} : \mathbb{R}^2 \rightarrow \mathbb{R}$, such that, for any set $A \in \mathbb{R}^2$, we have

$$P((X,Y) \in A) = \iint_A f_{XY}(x,y) dx dy \quad (5.15)$$

The function $f_{XY}(x,y)$ is called the **joint probability density function (PDF)** of X and Y .

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x,y) dx dy = 1$$

Marginal PDFs

s. 259

Marginal PDFs

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y) dy, \quad \text{for all } x,$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x,y) dx, \quad \text{for all } y.$$

Joint Cumulative Distribution Function (CDF)

s. 262

The **joint cumulative function** of two random variables X and Y is defined as

$$F_{XY}(x, y) = P(X \leq x, Y \leq y).$$

The joint CDF satisfies the following properties:

1. $F_X(x) = F_{XY}(x, \infty)$, for any x (marginal CDF of X);
2. $F_Y(y) = F_{XY}(\infty, y)$, for any y (marginal CDF of Y);
3. $F_{XY}(\infty, \infty) = 1$;
4. $F_{XY}(-\infty, y) = F_{XY}(x, -\infty) = 0$;
5. $P(x_1 < X \leq x_2, y_1 < Y \leq y_2) = F_{XY}(x_2, y_2) - F_{XY}(x_1, y_2) - F_{XY}(x_2, y_1) + F_{XY}(x_1, y_1)$;
6. if X and Y are independent, then $F_{XY}(x, y) = F_X(x)F_Y(y)$.

Continuous random variable and event A

s. 266-267

If X is a continuous random variable, and A is the event that $a < X < b$ (where possibly $b = \infty$ or $a = -\infty$), then

$$F_{X|A}(x) = \begin{cases} 1 & x > b \\ \frac{F_X(x) - F_X(a)}{F_X(b) - F_X(a)} & a \leq x < b \\ 0 & x < a \end{cases}$$

$$f_{X|A}(x) = \begin{cases} \frac{f_X(x)}{P(A)} & a \leq x < b \\ 0 & \text{otherwise} \end{cases}$$

$$E[X|A] = \int_{-\infty}^{\infty} x f_{X|A}(x) dx,$$

$$E[g(X)|A] = \int_{-\infty}^{\infty} g(x) f_{X|A}(x) dx,$$

$$\text{Var}(X|A) = E[X^2|A] - (E[X|A])^2$$

Two jointly continuous random variables X and Y

s. 269 + 271

For two jointly continuous random variables X and Y , we can define the following conditional concepts:

1. The conditional PDF of X given $Y = y$:

$$f_{X|Y}(x|y) = \frac{f_{XY}(x, y)}{f_Y(y)}$$

2. The conditional probability that $X \in A$ given $Y = y$:

$$P(X \in A|Y = y) = \int_A f_{X|Y}(x|y)dx$$

3. The conditional CDF of X given $Y = y$:

$$F_{X|Y}(x|y) = P(X \leq x|Y = y) = \int_{-\infty}^x f_{X|Y}(x|y)dx$$

For two jointly continuous random variables X and Y , we have:

1. Expected value of X given $Y = y$:

$$E[X|Y = y] = \int_{-\infty}^{\infty} x f_{X|Y}(x|y)dx$$

2. Conditional LOTUS:

$$E[g(X)|Y = y] = \int_{-\infty}^{\infty} g(x) f_{X|Y}(x|y)dx$$

3. Conditional variance of X given $Y = y$:

$$Var(X|Y = y) = E[X^2|Y = y] - (E[X|Y = y])^2$$

Independent Random Variables

s. 272

Two continuous random variables X and Y are independent if

$$f_{XY}(x, y) = f_X(x)f_Y(y), \quad \text{for all } x, y.$$

Equivalently, X and Y are independent if

$$F_{XY}(x, y) = F_X(x)F_Y(y), \quad \text{for all } x, y.$$

If X and Y are independent, we have

$$\begin{aligned} E[XY] &= E[X]E[Y], \\ E[g(X)h(Y)] &= E[g(X)]E[h(Y)]. \end{aligned}$$

Law of Total Probability

s. 275

Law of Total Probability:

$$P(A) = \int_{-\infty}^{\infty} P(A|X = x) f_X(x) dx \quad (5.16)$$

Law of Total Expectation:

$$\begin{aligned} E[Y] &= \int_{-\infty}^{\infty} E[Y|X = x] f_X(x) dx \quad (5.17) \\ &= E[E[Y|X]] \end{aligned}$$

Law of Total Variance:

$$\text{Var}(Y) = E[\text{Var}(Y|X)] + \text{Var}(E[Y|X]) \quad (5.18)$$

LOTUS for two continuous random variables

s. 278

LOTUS for two continuous random variables:

$$E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{XY}(x, y) dx dy \quad (5.19)$$

The Method of Transformations

s. 280-283

Theorem 5.1

Let X and Y be two jointly continuous random variables. Let $(Z, W) = g(X, Y) = (g_1(X, Y), g_2(X, Y))$, where $g: \mathbb{R}^2 \mapsto \mathbb{R}^2$ is a continuous one-to-one (invertible) function with continuous partial derivatives. Let $h = g^{-1}$, i.e., $(X, Y) = h(Z, W) = (h_1(Z, W), h_2(Z, W))$. Then Z and W are jointly continuous and their joint PDF, $f_{ZW}(z, w)$, for $(z, w) \in R_{ZW}$ is given by

$$f_{ZW}(z, w) = f_{XY}(h_1(z, w), h_2(z, w)) |J|,$$

where J is the Jacobian of h defined by

$$J = \det \begin{bmatrix} \frac{\partial h_1}{\partial z} & \frac{\partial h_1}{\partial w} \\ \frac{\partial h_2}{\partial z} & \frac{\partial h_2}{\partial w} \end{bmatrix} = \frac{\partial h_1}{\partial z} \cdot \frac{\partial h_2}{\partial w} - \frac{\partial h_2}{\partial z} \frac{\partial h_1}{\partial w}.$$

If X and Y are two jointly continuous random variables and $Z = X + Y$, then

$$f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(w, z-w)dw = \int_{-\infty}^{\infty} f_{XY}(z-w, w)dw.$$

If X and Y are also independent, then

$$\begin{aligned} f_Z(z) &= f_X(z) * f_Y(z) \\ &= \int_{-\infty}^{\infty} f_X(w)f_Y(z-w)dw = \int_{-\infty}^{\infty} f_Y(w)f_X(z-w)dw. \end{aligned}$$

Theorem 5.2

If $X \sim N(\mu_X, \sigma_X^2)$ and $Y \sim N(\mu_Y, \sigma_Y^2)$ are independent, then

$$X + Y \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2).$$

Covariance

s. 291-292

The **covariance** between X and Y is defined as

$$\text{Cov}(X, Y) = E[(X - EX)(Y - EY)] = E[XY] - (EX)(EY).$$

Lemma 5.3

The covariance has the following properties:

1. $\text{Cov}(X, X) = \text{Var}(X)$;
2. if X and Y are independent then $\text{Cov}(X, Y) = 0$;
3. $\text{Cov}(X, Y) = \text{Cov}(Y, X)$;
4. $\text{Cov}(aX, Y) = a\text{Cov}(X, Y)$;
5. $\text{Cov}(X + c, Y) = \text{Cov}(X, Y)$;
6. $\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$;
7. more generally,

$$\text{Cov}\left(\sum_{i=1}^m a_i X_i, \sum_{j=1}^n b_j Y_j\right) = \sum_{i=1}^m \sum_{j=1}^n a_i b_j \text{Cov}(X_i, Y_j).$$

Variance of a sum

s. 293

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y) \quad (5.21)$$

Correlation Coefficient

s. 294-295

$$\rho_{XY} = \rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Properties of the correlation coefficient:

1. $-1 \leq \rho(X, Y) \leq 1$;
2. if $\rho(X, Y) = 1$, then $Y = aX + b$, where $a > 0$;
3. if $\rho(X, Y) = -1$, then $Y = aX + b$, where $a < 0$;
4. $\rho(aX + b, cY + d) = \rho(X, Y)$ for $a, c > 0$.

Definition 5.2

Consider two random variables X and Y :

- If $\rho(X, Y) = 0$, we say that X and Y are **uncorrelated**.
- If $\rho(X, Y) > 0$, we say that X and Y are **positively** correlated.
- If $\rho(X, Y) < 0$, we say that X and Y are **negatively** correlated.

If X and Y are uncorrelated, then

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

More generally, if $X_1, X_2, \dots, X_n$ are pairwise uncorrelated, i.e., $\rho(X_i, X_j) = 0$ when $i \neq j$, then

$$\text{Var}(X_1 + X_2 + \dots + X_n) = \text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n).$$

Bivariate normal/Jointly normal

s. 297-303

Definition 5.3

Two random variables X and Y are said to be **bivariate normal**, or **jointly normal**, if $aX + bY$ has a normal distribution for all $a, b \in \mathbb{R}$.

Two random variables X and Y are said to have the **standard bivariate normal distribution with correlation coefficient** ρ if their joint PDF is given by

$$f_{XY}(x, y) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}[x^2 - 2\rho xy + y^2]\right\},$$

where $\rho \in (-1, 1)$. If $\rho = 0$, then we just say X and Y have the standard bivariate normal distribution.

Definition 5.4

Two random variables X and Y are said to have a **bivariate normal distribution** with parameters $\mu_X, \sigma_X^2, \mu_Y, \sigma_Y^2$, and ρ , if their joint PDF is given by

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \cdot \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\frac{(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y}\right]\right\} \quad (5.24)$$

where $\mu_X, \mu_Y \in \mathbb{R}$, $\sigma_X, \sigma_Y > 0$ and $\rho \in (-1, 1)$ are all constants.

Theorem 5.4

Suppose X and Y are jointly normal random variables with parameters $\mu_X, \sigma_X^2, \mu_Y, \sigma_Y^2$, and ρ . Then, given $X = x$, Y is normally distributed with

$$E[Y|X = x] = \mu_Y + \rho\sigma_Y\frac{x - \mu_X}{\sigma_X},$$

$$Var(Y|X = x) = (1 - \rho^2)\sigma_Y^2.$$

Chapter 6 - Multiple Random Variables

Independent and identically distributed

s. 321

Definition 6.1. Random variables $X_1, X_2, \dots, X_n$ are said to be **independent and identically distributed (i.i.d.)** if they are *independent*, and they have the *same marginal distributions*:

$$F_{X_1}(x) = F_{X_2}(x) = \dots = F_{X_n}(x), \text{ for all } x \in \mathbb{R}.$$

Sums of Random Variables

s. 322

$$\text{Var} \left(\sum_{i=1}^n X_i \right) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j)$$

If $X_1, X_2, \dots, X_n$ are independent,
$$\text{Var} \left(\sum_{i=1}^n X_i \right) = \sum_{i=1}^n \text{Var}(X_i).$$

Moment Generating Functions

s. 335

The moment generating function (MGF) of a random variable X is a function $M_X(s)$ defined as

$$M_X(s) = E[e^{sX}].$$

We say that MGF of X exists, if there exists a positive constant a such that $M_X(s)$ is finite for all $s \in [-a, a]$.

Finding Moments from MGF

s. 237

We can obtain all moments of X^k from its MGF:

$$M_X(s) = \sum_{k=0}^{\infty} E[X^k] \frac{s^k}{k!},$$

$$E[X^k] = \frac{d^k}{ds^k} M_X(s) \big|_{s=0}.$$

Sum of Independent Random Variables

s. 329

If $X_1, X_2, \dots, X_n$ are n independent random variables, then

$$M_{X_1+X_2+\dots+X_n}(s) = M_{X_1}(s)M_{X_2}(s) \cdots M_{X_n}(s).$$

Correlation and Covariance Matrix

s. 333

Correlation matrix of $\mathbf{X}$:

$$\mathbf{R}_X = \mathbf{E}[\mathbf{X}\mathbf{X}^T]$$

Covariance matrix of $\mathbf{X}$:

$$\mathbf{C}_X = \mathbf{E}[(\mathbf{X} - \mathbf{E}\mathbf{X})(\mathbf{X} - \mathbf{E}\mathbf{X})^T] = \mathbf{R}_X - \mathbf{E}\mathbf{X}\mathbf{E}\mathbf{X}^T$$

Chapter 8 - Statistical Inference I: Classical Methods

Random sample

s. 425-426

The collection of random variables $X_1, X_2, X_3, \dots, X_n$ is said to be a random sample of size n if they are independent and identically distributed (i.i.d)

1. $X_1, X_2, X_3, \dots, X_n$ are independent random variables, and
2. They have the same distribution, $F_{X_1}(x) = F_{X_2}(x) = \dots = F_{X_n}(x)$, for all $x \in \mathbb{R}$

Some properties of Random Samples

$$EX_i = EX = \mu < \infty$$

$$0 < Var(X_i) = Var(X) = \sigma^2 < \infty$$

Sample mean

s. 426

$$\bar{X} = \frac{X_1 + \dots + X_n}{n}$$

Properties of the sample mean

$$E\bar{X} = \mu$$

$$Var(\bar{X}) = \frac{\sigma^2}{n}$$

$$\lim_{n \rightarrow \infty} P(|\bar{X} - \mu| \geq \varepsilon) = 0 \text{ (Weak Law of Large Numbers)}$$

$$Z_n = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{X_1 + X_2 + \dots + X_n - n\mu}{\sqrt{n}\sigma} \text{ (Central Limit Theorem)}$$

Theorem 8.1

s. 427

Theorem 8.1

Let $X_1, X_2, \dots, X_n$ be a random sample from a continuous distribution with CDF $F_X(x)$ and PDF $f_X(x)$. Let $X_{(1)}, X_{(2)}, \dots, X_{(n)}$ be the order statistics of $X_1, X_2, X_3, \dots, X_n$. Then the CDF and PDF of $X_{(i)}$ are given by

$$f_{X_{(i)}}(x) = \frac{n!}{(i-1)!(n-i)!} f_X(x) [F_X(x)]^{i-1} [1 - F_X(x)]^{n-i},$$

$$F_{X_{(i)}}(x) = \sum_{k=i}^n \binom{n}{k} [F_X(x)]^k [1 - F_X(x)]^{n-k}.$$

Also, the joint PDF of $X_{(1)}, X_{(2)}, \dots, X_{(n)}$ is given by

$$f_{X_{(1)}, \dots, X_{(n)}}(x_1, x_2, \dots, x_n) = \begin{cases} n! f_X(x_1) f_X(x_2) \cdots f_X(x_n) & \text{for } x_1 \leq x_2 \leq x_3 \cdots \leq x_n \\ 0 & \text{otherwise} \end{cases}$$

Bias

s. 429

Let $\hat{\theta} = h(X_1, X_2, \dots, X_n)$ be a point estimator for θ . The bias of point estimator is $\hat{\theta}$ is defined by

$$B(\hat{\theta}) = E[\hat{\theta}] - \theta$$

In general, a point estimator is a function of the random sample $\hat{\theta} = h(X_1, X_2, \dots, X_n)$ that is used to estimate an unknown quantity. (s. 425)

Unbiased

s. 429

Let $\hat{\theta} = h(X_1, X_2, \dots, X_n)$ be a point estimator for θ . We say that $\hat{\theta}$ is an unbiased estimator of θ if

$$B(\hat{\theta}) = 0, \text{ for all possible values of } \theta$$

Mean squared error (MSE)

s. 430-431

The mean squared error (MSE) of a point estimator $\hat{\theta}$, shown by $MSE(\hat{\theta})$, is defined as

$$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$$

If $\hat{\theta}$ is a point estimator for θ ,

$$MSE(\hat{\theta}) = Var(\hat{\theta}) + B(\hat{\theta})^2$$

Where $B(\hat{\theta})^2 = E[\hat{\theta}] - \theta$ is the bias of $\hat{\theta}$.

Let $\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_n, \dots$, be a sequence of point estimators of θ . We say that $\hat{\theta}_n$ is a consistent estimator of θ , if

$$\lim_{n \rightarrow \infty} P(|\hat{\theta} - \theta| \geq \varepsilon) = 0$$

Sample variance

s. 434

$$S^2 = \frac{1}{n-1} \sum_{k=1}^n (X_k - \bar{X})^2$$

Sample standard deviation

s. 434

$$S = \sqrt{S^2}$$

Likelihood function

s. 437

Let $X_1, X_2, X_3, \dots, X_n$ be a random sample from a distribution with a parameter θ . Suppose that we have observed $X_1 = x_1, X_2 = x_2, \dots, X_n = x_n$.

1. If X_i 's are discrete, then the **likelihood function** is defined as

$$L(x_1, x_2, \dots, x_n; \theta) = P_{X_1 X_2 \dots X_n}(x_1, x_2, \dots, x_n; \theta).$$

2. If X_i 's are jointly continuous, then the likelihood function is defined as

$$L(x_1, x_2, \dots, x_n; \theta) = f_{X_1 X_2 \dots X_n}(x_1, x_2, \dots, x_n; \theta).$$

In some problems, it is easier to work with the **log likelihood function** given by

$$\ln L(x_1, x_2, \dots, x_n; \theta).$$

Maximum Likelihood Estimator (MLE)

s. 440

The Maximum Likelihood Estimator (MLE)

Let $X_1, X_2, X_3, \dots, X_n$ be a random sample from a distribution with a parameter θ . Given that we have observed $X_1 = x_1, X_2 = x_2, \dots, X_n = x_n$, a maximum likelihood estimate of θ , shown by $\hat{\theta}_{ML}$ is a value of θ that maximizes the likelihood function

$$L(x_1, x_2, \dots, x_n; \theta).$$

A maximum likelihood estimator (MLE) of the parameter θ , shown by $\hat{\Theta}_{ML}$ is a random variable $\hat{\Theta}_{ML} = \hat{\Theta}_{ML}(X_1, X_2, \dots, X_n)$ whose value when $X_1 = x_1, X_2 = x_2, \dots, X_n = x_n$ is given by $\hat{\theta}_{ML}$.

It is worth noting that technically, we need to look at the second derivatives and endpoints to make sure that the values that we obtained above are the maximizing values. For this example, it turns out that the obtained values are indeed the maximizing values.

Asymptotic Properties of MLEs

s. 443

Asymptotic Properties of MLEs

Let $X_1, X_2, X_3, \dots, X_n$ be a random sample from a distribution with a parameter θ . Let $\hat{\Theta}_{ML}$ denote the maximum likelihood estimator (MLE) of θ . Then, under some mild regularity conditions,

1. $\hat{\Theta}_{ML}$ is asymptotically consistent, i.e.,

$$\lim_{n \rightarrow \infty} P(|\hat{\Theta}_{ML} - \theta| > \epsilon) = 0.$$

2. $\hat{\Theta}_{ML}$ is asymptotically unbiased, i.e.,

$$\lim_{n \rightarrow \infty} E[\hat{\Theta}_{ML}] = \theta.$$

3. As n becomes large, $\hat{\Theta}_{ML}$ is approximately a normal random variable. More precisely, the random variable

$$\frac{\hat{\Theta}_{ML} - \theta}{\sqrt{\text{Var}(\hat{\Theta}_{ML})}}$$

converges in distribution to $N(0, 1)$.

Interval Estimation

s. 449

Interval Estimation

Let $X_1, X_2, X_3, \dots, X_n$ be a random sample from a distribution with a parameter θ that is to be estimated. An **interval estimator** with **confidence level** $1 - \alpha$ consists of two estimators $\hat{\Theta}_l(X_1, X_2, \dots, X_n)$ and $\hat{\Theta}_h(X_1, X_2, \dots, X_n)$ such that

$$P\left(\hat{\Theta}_l \leq \theta \text{ and } \hat{\Theta}_h \geq \theta\right) \geq 1 - \alpha,$$

for every possible value of θ . Equivalently, we say that $[\hat{\Theta}_l, \hat{\Theta}_h]$ is a $(1 - \alpha)100\%$ **confidence interval** for θ .

Pivotal Quantity

s. 453

Pivotal Quantity

Let $X_1, X_2, X_3, \dots, X_n$ be a random sample from a distribution with a parameter θ that is to be estimated. The random variable Q is said to be a *pivot* or a *pivotal quantity*, if it has the following properties:

1. It is a function of the observed data $X_1, X_2, X_3, \dots, X_n$ and the unknown parameter θ , but it does not depend on any other unknown parameters:

$$Q = Q(X_1, X_2, \dots, X_n, \theta).$$

2. The probability distribution of Q does not depend on θ or any other unknown parameters.

Find Confidence interval

s. 455-460

Assumptions: A random sample $X_1, X_2, X_3, \dots, X_n$ is given from a distribution with known variance $\text{Var}(X_i) = \sigma^2 < \infty$; n is large.

Parameter to be Estimated: $\theta = EX_i$.

Confidence Interval: $\left[\bar{X} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}\right]$ is approximately a $(1 - \alpha)100\%$ confidence interval for θ .

Assumptions: A random sample $X_1, X_2, X_3, \dots, X_n$ is given from a $Bernoulli(\theta)$; n is large.

Parameter to be Estimated: θ

Confidence Interval: $\left[\bar{X} - \frac{z_{\frac{\alpha}{2}}}{2\sqrt{n}}, \bar{X} + \frac{z_{\frac{\alpha}{2}}}{2\sqrt{n}} \right]$ is approximately a $(1 - \alpha)100\%$ confidence interval for θ . This is a conservative confidence interval as it is obtained using an upper bound for σ .

Assumptions: A random sample $X_1, X_2, X_3, \dots, X_n$ is given from a $Bernoulli(\theta)$; n is large.

Parameter to be Estimated: θ

Confidence Interval: $\left[\bar{X} - z_{\frac{\alpha}{2}} \sqrt{\frac{\bar{X}(1-\bar{X})}{n}}, \bar{X} + z_{\frac{\alpha}{2}} \sqrt{\frac{\bar{X}(1-\bar{X})}{n}} \right]$ is approximately a $(1 - \alpha)100\%$ confidence interval for θ .

Assumptions: A random sample $X_1, X_2, X_3, \dots, X_n$ is given from a distribution with unknown variance $\text{Var}(X_i) = \sigma^2 < \infty$; n is large.

Parameter to be Estimated: $\theta = EX_i$.

Confidence Interval: If S is the sample standard deviation

$$S = \sqrt{\frac{1}{n-1} \sum_{k=1}^n (X_k - \bar{X})^2} = \sqrt{\frac{1}{n-1} \left(\sum_{k=1}^n X_k^2 - n\bar{X}^2 \right)},$$

then the interval

$$\left[\bar{X} - z_{\frac{\alpha}{2}} \frac{S}{\sqrt{n}}, \bar{X} + z_{\frac{\alpha}{2}} \frac{S}{\sqrt{n}} \right]$$

is approximately a $(1 - \alpha)100\%$ confidence interval for θ .

Chi-Squared Distribution

s. 462-463

Definition 8.1. If $Z_1, Z_2, \dots, Z_n$ are independent standard normal random variables, the random variable Y defined as

$$Y = Z_1^2 + Z_2^2 + \dots + Z_n^2$$

is said to have a *chi-squared* distribution with n degrees of freedom shown by

$$Y \sim \chi^2(n).$$

Properties:

1. The chi-squared distribution is a special case of the gamma distribution. More specifically,

$$Y \sim \text{Gamma}\left(\frac{n}{2}, \frac{1}{2}\right).$$

Thus,

$$f_Y(y) = \frac{1}{2^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)} y^{\frac{n}{2}-1} e^{-\frac{y}{2}}, \quad \text{for } y > 0.$$

2. $EY = n$, $\text{Var}(Y) = 2n$.
3. For any $p \in [0, 1]$ and $n \in \mathbb{N}$, we define $\chi_{p,n}^2$ as the real value for which

$$P(Y > \chi_{p,n}^2) = p,$$

tr

Theorem 8.3. Let $X_1, X_2, \dots, X_n$ be i.i.d. $N(\mu, \sigma^2)$ random variables. Also, let S^2 be the sample variance for this random sample. Then, the random variable Y defined as

$$Y = \frac{(n-1)S^2}{\sigma^2} = \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \bar{X})^2$$

has a chi-squared distribution with $n-1$ degrees of freedom, i.e., $Y \sim \chi^2(n-1)$. Moreover, $\bar{X}$ and S^2 are independent random variables.

The t-Distribution

s. 465-466

Definition 8.2. Let $Z \sim N(0, 1)$, and $Y \sim \chi^2(n)$, where $n \in \mathbb{N}$. Also assume that Z and Y are independent. The random variable T defined as

$$T = \frac{Z}{\sqrt{Y/n}}$$

is said to have a t -distribution with n degrees of freedom shown by

$$T \sim T(n).$$

Properties:

1. The t -distribution has a bell-shaped PDF centered at 0, but its PDF is more spread out than the normal PDF (Figure 8.7).
2. $ET = 0$, for $n > 0$. But ET , is undefined for $n = 1$.
3. $\text{Var}(T) = \frac{n}{n-2}$, for $n > 2$. But, $\text{Var}(T)$ is undefined for $n = 1, 2$.
4. As n becomes large, the t density approaches the standard normal PDF. More formally, we can write

$$T(n) \xrightarrow{d} N(0, 1).$$

5. For any $p \in [0, 1]$ and $n \in \mathbb{N}$, we define $t_{p,n}$ as the real value for which

$$P(T > t_{p,n}) = p.$$

Since the t -distribution has a symmetric PDF, we have

$$t_{1-p,n} = -t_{p,n}.$$

Theorem 8.4. Let $X_1, X_2, \dots, X_n$ be i.i.d. $N(\mu, \sigma^2)$ random variables. Also, let S^2 be the sample variance for this random sample. Then, the random variable T defined as

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has a t -distribution with $n - 1$ degrees of freedom, i.e., $T \sim T(n - 1)$.

Confidence Intervals for the mean of Normal Random Variables

s. 466-469

Assumptions: A random sample $X_1, X_2, X_3, \dots, X_n$ is given from a $N(\mu, \sigma^2)$ distribution, where $\text{Var}(X_i) = \sigma^2$ is known.

Parameter to be Estimated: $\mu = EX_i$.

Confidence Interval: $\left[\bar{X} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \right]$ is a $(1 - \alpha)100\%$ confidence interval for μ .

Assumptions: A random sample $X_1, X_2, X_3, \dots, X_n$ is given from a $N(\mu, \sigma^2)$ distribution, where $\mu = EX_i$ and $\text{Var}(X_i) = \sigma^2$ are unknown.

Parameter to be Estimated: $\mu = EX_i$.

Confidence Interval: $\left[\bar{X} - t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}}, \bar{X} + t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}} \right]$ is a $(1 - \alpha)$ confidence interval for μ .

Assumptions: A random sample $X_1, X_2, X_3, \dots, X_n$ is given from a $N(\mu, \sigma^2)$ distribution, where $\mu = EX_i$ and $\text{Var}(X_i) = \sigma^2$ are unknown.

Parameter to be Estimated: $\text{Var}(X_i) = \sigma^2$.

Confidence Interval: $\left[\frac{(n-1)S^2}{\chi_{\frac{\alpha}{2}, n-1}^2}, \frac{(n-1)S^2}{\chi_{1-\frac{\alpha}{2}, n-1}^2} \right]$ is a $(1 - \alpha)100\%$ confidence interval for σ^2 .

Hypothesis testing

s. 474 + 477

H_0 (the **null** hypothesis): $\theta \in S_0$.

H_1 (the **alternative** hypothesis): $\theta \in S_1$.

Definition 8.3. Let $X_1, X_2, \dots, X_n$ be a random sample of interest. A **statistic** is a real-valued function of the data. For example, the sample mean, defined as

$$W(X_1, X_2, \dots, X_n) = \frac{X_1 + X_2 + \dots + X_n}{n},$$

is a statistic. A **test statistic** is a statistic based on which we build our test.

Two-sided hypothesis testing $H_0: \mu = \mu_0, H_1: \mu \neq \mu_0$

s. 485

Table 8.2: Two-sided hypothesis testing for the mean: $H_0: \mu = \mu_0, H_1: \mu \neq \mu_0$.

Case	Test Statistic	Acceptance Region
$X_i \sim N(\mu, \sigma^2), \sigma$ known	$W = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$	$ W \leq z_{\frac{\alpha}{2}}$
n large, X_i non-normal	$W = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$	$ W \leq z_{\frac{\alpha}{2}}$
$X_i \sim N(\mu, \sigma^2), \sigma$ unknown	$W = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$	$ W \leq t_{\frac{\alpha}{2}, n-1}$

One-sided hypothesis testing for the mean $H_0: \mu \geq \mu_0, H_1: \mu < \mu_0$

s. 488

Table 8.4: One-sided hypothesis testing for the mean: $H_0: \mu \geq \mu_0, H_1: \mu < \mu_0$.

Case	Test Statistic	Acceptance Region
$X_i \sim N(\mu, \sigma^2), \sigma$ known	$W = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$	$W \geq -z_{\alpha}$
n large, X_i non-normal	$W = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$	$W \geq -z_{\alpha}$
$X_i \sim N(\mu, \sigma^2), \sigma$ unknown	$W = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$	$W \geq -t_{\alpha, n-1}$

One-sided hypothesis testing for the mean $H_0: \mu \leq \mu_0, H_1: \mu > \mu_0$

s. 488

Table 8.3: One-sided hypothesis testing for the mean: $H_0: \mu \leq \mu_0, H_1: \mu > \mu_0$.

Case	Test Statistic	Acceptance Region
$X_i \sim N(\mu, \sigma^2), \sigma \text{ known}$	$W = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$	$W \leq z_\alpha$
$n \text{ large, } X_i \text{ non-normal}$	$W = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$	$W \leq z_\alpha$
$X_i \sim N(\mu, \sigma^2), \sigma \text{ unknown}$	$W = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$	$W \leq t_{\alpha, n-1}$

P-value

s. 488

P-value is the lowest significance level α that results in rejecting the null hypothesis.

Computing P-Values

Consider a hypothesis test for choosing between H_0 and H_1 . Let W be the test statistic, and w_1 be the observed value of W .

1. Assume H_0 is true.
2. The P -value is $P(\text{type I error})$ when the test threshold c is chosen to be $c = w_1$.

Likelihood Ratio Test for Simple Hypotheses

s. 491

Likelihood Ratio Test for Simple Hypotheses

Let $X_1, X_2, X_3, \dots, X_n$ be a random sample from a distribution with a parameter θ . Suppose that we have observed $X_1 = x_1, X_2 = x_2, \dots, X_n = x_n$. To decide between two simple hypotheses

$$H_0: \theta = \theta_0,$$

$$H_1: \theta = \theta_1,$$

we define

$$\lambda(x_1, x_2, \dots, x_n) = \frac{L(x_1, x_2, \dots, x_n; \theta_0)}{L(x_1, x_2, \dots, x_n; \theta_1)}.$$

To perform a **likelihood ratio test (LRT)**, we choose a constant c . We reject H_0 if $\lambda < c$ and accept it if $\lambda \geq c$. The value of c can be chosen based on the desired α .

Likelihood Ratio Tests

Let $X_1, X_2, X_3, \dots, X_n$ be a random sample from a distribution with a parameter θ . Suppose that we have observed $X_1 = x_1, X_2 = x_2, \dots, X_n = x_n$. Define

$$\lambda(x_1, x_2, \dots, x_n) = \frac{\sup\{L(x_1, x_2, \dots, x_n; \theta) : \theta \in S_0\}}{\sup\{L(x_1, x_2, \dots, x_n; \theta) : \theta \in S\}}.$$

To perform a **likelihood ratio test (LRT)**, we choose a constant c in $[0, 1]$. We reject H_0 if $\lambda < c$ and accept it if $\lambda \geq c$. The value of c can be chosen based on the desired α .

Simple Linear Regression

s. 501

Given the observations $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, we can write the regression line as

$$\hat{y} = \beta_0 + \beta_1 x.$$

We can estimate β_0 and β_1 as

$$\begin{aligned}\hat{\beta}_1 &= \frac{s_{xy}}{s_{xx}}, \\ \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x},\end{aligned}$$

where

$$\begin{aligned}s_{xx} &= \sum_{i=1}^n (x_i - \bar{x})^2, \\ s_{xy} &= \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}).\end{aligned}$$

For each x_i , the **fitted value** $\hat{y}_i$ is obtained by

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i.$$

The quantities

$$e_i = y_i - \hat{y}_i$$

are called the **residuals**.

Coefficient of Determination

s. 505

Coefficient of Determination

For the observed data pairs, $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, we define **coefficient of determination**, r^2 as

$$r^2 = \frac{s_{xy}^2}{s_{xx} s_{yy}},$$

where

$$s_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2, \quad s_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2, \quad s_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}).$$

We have $0 \leq r^2 \leq 1$. Larger values of r^2 generally suggest that our linear model

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

is a good fit for the data.

Chapter 11 - Some important random processes

Markov Chains

Discrete-Time Markov Chains

s. 631

Consider the random process $\{X_n, n = 0, 1, 2, \dots\}$, where $R_{X_i} = S \subseteq \{0, 1, 2, \dots\}$. This process is a Markov chain if

$$P(X_{m+1} = j | X_m = i, X_{m-1} = i_{m-1}, \dots, X_0 = i_0) = P(X_{m+1} = j | X_m = i), \quad \text{for all } m, j, i, i_0, i_1, \dots, i_{m-1}$$

If the number of states is finite, $S = \{0, 1, 2, \dots, r\}$, we call it *finite Markov chain*.

Transition probabilities

s. 631

$$p_{ij} = P(X_{m+1} = j | X_m = i)$$

With other words, if the process is in state i , it will next make a transition to state j with probability p_{ij} .

State Transition Matrix

s. 631

Assuming the states are $1, 2, \dots, r$

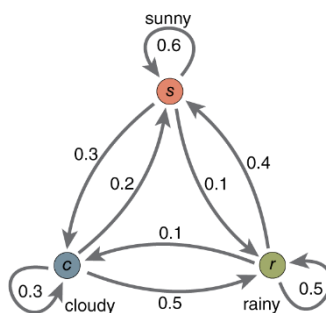
$$P = \begin{bmatrix} p_{11} & \cdots & p_{1r} \\ \vdots & \ddots & \vdots \\ p_{r1} & \cdots & p_{rr} \end{bmatrix}$$

Note that $p_{ij} \geq 0$, and for all i

$$\sum_{k=1}^r p_{ik} = \sum_{k=1}^r P(X_{m+1} = k | X_m = i) = 1$$

State Transition Diagram

s. 632



State Probability Distributions

s. 634

$$\pi^{(n+1)} = \pi^{(n)} P, \quad \text{for } n = 0, 1, 2, \dots$$

$$\pi^{(n)} = \pi^{(0)} P^n, \quad \text{for } n = 0, 1, 2, \dots$$

$\pi^{(n)} = \pi^{(0)} p^n$ is used to find the probability that the system is in by some started state at time n .

The chapman-Kolmogorov equation

s. 636

$$p_{ij}^{(m+n)} = P(X_{m+n} = j | X_0 = i) = \sum_{k \in S} P_{ik}^{(m)} P_{kj}^{(n)}$$

 n -step transition matrix

s. 636

$$P^{(n)} = P^n, \quad \text{for } n = 1, 2, 3, \dots$$

Accessible

s. 637

We say that state j is accessible from state i , written as $i \rightarrow j$, if $p_{ij}^{(n)} > 0$ for some n . We assume every state is accessible from itself since $p_{ii}^{(n)} = 1$

Communicate

s. 637

Two states i and j are said to communicate, written as $i \leftrightarrow j$, if they are accessible from each other.

$i \leftrightarrow j$ means $i \rightarrow j$ and $j \rightarrow i$.

Markov chain can be partitioned into communicating classes such that only members of the same class communicate with each other. The only way, two state, i and j , can be in same class is if only, they communicate with each other, $i \leftrightarrow j$.

Irreducible

s. 638

A Markov chain is said to be irreducible if all states communicate with each other.

With other words, if you can get from i to j and vice versa, then they are irreducible. But if you only can get from i to j and not the other way around, then they are not irreducible.

Recurrent and transient state

s. 639

For any state i , we define

$$f_{ii} = P(X_n = i, \text{ for some } n \geq 1 | X_0 = i)$$

State i is recurrent if $f_{ii} = 1$, and it is transient if $f_{ii} < 1$.

Recurrent states are if the Markov chain might stay in some class, jump from and back to other classes, but at some point, it will leave that class and it will never return to that class again. Look at Figure 11.10 s. 638. Class 4 is recurrent state. All other states in that Markov chain are transient.

Consider a discrete-time Markov chain. Let V be the total number of visits to state i .

If i is a recurrent state, then

$$P(V = \infty | X_0 = i) = 1$$

If i is a transient state, then

$$V | X_0 = i \sim \text{Geometric}(1 - f_{ii})$$

This means, that if i is a recurrent state, then the chain will return to state i any time it leaves that state. The chain will visit state i an infinite number of times.

If i is a transient state, then the chain will return with the probability $f_{ii} < 1$.

Periodic and aperiodic

s. 640

The period of a state i is the largest integer d satisfying the following property:

$p_{ii}^{(n)} = 0$, whenever n is not divisible by d . The period of i is shown by $d(i)$. if $p_{ii}^{(n)} > 0$, for all $n > 0$, then we let $d(i) = \infty$.

If $d(i) > 1$, then state i is periodic.

If $d(i) = 1$, then state i is aperiodic.

Periodic is the length we move from state i and then return to state i .

A class is said to be aperiodic if its states are aperiodic. Same goes for the whole Markov chain.

Absorption Probabilities

s. 644

Consider a finite Markov chain $\{X_n, n = 0, 1, 2, \dots\}$ with state space $S = \{0, 1, 2, \dots, r\}$. Suppose that all states are either absorbing or transient. Let $l \in S$ be an absorbing state. Define

$$a_i = P(\text{absorption in } l | X_0 = i), \quad \text{for all } i \in S.$$

By the above definition, we have $a_l = 1$, and $a_j = 0$ if j is any other absorbing state. To find the unknown values of a_i 's, we can use the following equations

$$a_i = \sum_k a_k p_{ik}, \quad \text{for } i \in S.$$

Mean Hitting Time

s. 645

Consider a finite Markov chain $\{X_n, n = 0, 1, 2, \dots\}$ with state space $S = \{0, 1, 2, \dots, r\}$. Let $A \subset S$ be a set of states. Let T be the first time the chain visits a state in A . For all $i \in S$, define

$$t_i = E[T | X_0 = i].$$

By the above definition, we have $t_j = 0$, for all $j \in A$. To find the unknown values of t_i 's, we can use the following equations

$$t_i = 1 + \sum_k t_k p_{ik}, \quad \text{for } i \in S - A.$$

Mean Return Times

s. 647

Consider a finite irreducible Markov chain $\{X_n, n = 0, 1, 2, \dots\}$ with state space $S = \{0, 1, 2, \dots, r\}$. Let $l \in S$ be a state. Let r_l be the **mean return time** to state l . Then

$$r_l = 1 + \sum_k t_k p_{lk},$$

where t_k is the expected time until the chain hits state l given $X_0 = k$. Specifically,

$$\begin{aligned} t_l &= 0, \\ t_k &= 1 + \sum_j t_j p_{kj}, \quad \text{for } k \neq l. \end{aligned}$$

Limiting Distributions

s. 649

The probability distribution $\pi = [\pi_0, \pi_1, \pi_2, \dots]$ is called the **limiting distribution** of the Markov chain X_n if

$$\pi_j = \lim_{n \rightarrow \infty} P(X_n = j | X_0 = i)$$

for all $i, j \in S$, and we have

$$\sum_{j \in S} \pi_j = 1.$$

Finite Markov Chains

s. 653

Consider a finite Markov chain $\{X_n, n = 0, 1, 2, \dots\}$ where $X_n \in S = \{0, 1, 2, \dots, r\}$. Assume that the chain is irreducible and aperiodic. Then,

1. The set of equations

$$\begin{aligned} \pi &= \pi P, \\ \sum_{j \in S} \pi_j &= 1 \end{aligned}$$

has a unique solution.

2. The unique solution to the above equations is the limiting distribution of the Markov chain, i.e.,

$$\pi_j = \lim_{n \rightarrow \infty} P(X_n = j | X_0 = i),$$

for all $i, j \in S$.

3. We have

$$r_j = \frac{1}{\pi_j}, \quad \text{for all } j \in S,$$

where r_j is the mean return time to state j .